Project Proposal

Smart Language Learner Platform

NUS-ISS MTech Software Engineering — Capstone Project

Sponsor: Darryl Ng, NUS-ISS

Project Objectives

This capstone project aims to deliver the Smart Language Learner Platform — a deployable, AI-assisted learning system supporting composition writing, automated grading, and personalised feedback for language learners.

The project objectives are:

- Apply microservices architecture, domain-driven design, and cloud-native deployment practices to build a scalable, production-grade platform.
- Integrate AI and LLM capabilities to power composition generation, grading, and writing refinement, demonstrating advanced applied AI engineering.
- Demonstrate mastery of the full software engineering lifecycle, from requirements through architecture, implementation, testing, security hardening, and DevOps.
- Deliver a system of genuine value to the sponsoring organisation, supporting students, teachers, and self-learners across English and Chinese composition.
- Provide the sponsoring organisation early access to emerging AI-driven education technology, with potential for commercial extension.

The Team

The project is delivered by a team of five part-time MTech Software Engineering students (SE33 cohort), each bringing professional engineering experience:

Name	Role / Employer	Responsibilities
Yasin	Technical Team Lead at Deloitte	Technical Lead, DevOps & Infrastructure
Ming Wei	Back-end Software Engineer at DBS	Backend Microservices, AI Integration
Svetha	Software Engineer at Bank of America	Backend Microservices, Database Design
Jonathan	UX Engineer at NUS	Frontend, UX Design, Prototyping, Testing
Ma Xin	Client Software Developer at Tencent	Frontend, DevOps, QA

Key Project Dates

Milestone	Date
Project Kickoff	6 Apr 2026
Phase 0 Complete (Infrastructure Ready)	3 May 2026
Sprint 1 Review	31 May 2026
Mid-Project Review	End July 2026
Final Submission & Handover	End October 2026

Project Phases

The project is structured across seven phases, spanning infrastructure setup through final handover:

Phase	Period	Key Deliverables
Phase 0	6 Apr – 3 May	Repo setup, Infra setup (EKS, Jenkins, Terraform), DB schema, architecture design
Sprint 1	4 May – 31 May	Auth backend + UI, Document Upload Pipeline (PDF/DOCX/TXT)
Sprint 2	1 Jun – 28 Jun	AI Writing Engine (LLM + framework selector), Rubric/Framework Engine, Composition editor UI, Admin rubric UI, API gateway + microservices structure
Sprint 3	29 Jun – 26 Jul	AI Writing Engine (LLM + framework selector), Rubric/Framework Engine, AI Evaluation Engine (scoring + explainable feedback), Evaluation result review UI
Sprint 4	27 Jul – 23 Aug	unit + integration tests, Security hardening, stress testing
Sprint 5	24 Aug – 20 Sep	Chinese composition support, Show-Dont-Tell detection engine
Sprint 6	21 Sep – 18 Oct	Bug fixes, Final presentation prep, demo videos, report, code freeze
Buffer	19 – 30 Oct	Address presentation feedback, sponsor sign-off, final submission

Core system domains covered across all phases:

- User Management & Authentication
- Composition Writing Engine (AI-assisted generation)
- Evaluation & Grading Engine (rule-based + AI hybrid)
- Rubric & Framework Engine
- Knowledge Teaching Module (idioms, writing techniques)
- Document Processing Pipeline (PDF, DOCX, TXT upload)
- Refinement Writing Engine
- Analytics & Feedback System
- Admin & Content Management
- Security, Performance & DevOps

Project Conduct

The project will be delivered using Agile methodology, with work structured into 4-week sprints. This allows for continuous feedback, adaptability, and regular visibility of progress.

Sprint Ceremonies

Each sprint will include:

- A clearly defined set of sprint objectives and user stories
- Design, development, and testing of agreed features
- A sprint review session with the sponsor

At the end of each sprint, a one-hour review session will be held with the sponsor to:

- Demonstrate completed functionality
- Gather feedback and validate technical direction
- Clarify requirements and reprioritize for the next sprint
- Review sprint burndown charts and velocity

Technical Information

The platform will be designed as a cloud-hosted, microservices-based web application using production-grade open-source technologies.

Architecture Overview

Layer	Technology Choices
Frontend	- ReactJS: Component-based JavaScript framework. - React Router DOM: Routing library enabling navigation in React. - Axios: HTTP client for asynchronous API requests. - Tailwind CSS: Utility-first CSS framework. - Shad/cn: Collection of headless UI components. - React Context: State management for sharing data globally. - React i18next: Localization framework for managing translations and multi-languages.
API Gateway	Spring Cloud Gateway: request routing, rate limiting, auth enforcement
Backend Services	Java Spring Boot microservices (one per domain): DDD-aligned service boundaries
AI Integration	OpenAI / LLM API: composition generation, grading, refinement prompts
Data Store	PostgreSQL with Liquibase migrations: per-service databases
Infrastructure	Docker + Kubernetes on AWS EKS, provisioned via Terraform
CI/CD	Jenkins pipelines — build, test, containerize, deploy per service
Security	JWT authentication, HTTPS, RBAC, secrets management, OWASP practices
Observability	Observability / Monitoring through Logging, metrics, and distributed tracing
Testing	Unit, integration, and API testing strategy
Service Communication	REST-based inter-service communication (synchronous)

Support and Handover

The system will be designed for confident ownership and operation by NUS-ISS, Prof. Darryl Ng after project completion, with no long-term dependency on the development team.

Handover Deliverables

At project completion, the team will provide:

- Full source code in a well-documented GitHub mono-repository
- System and microservices architecture documentation with diagrams
- CI/CD pipeline documentation and Terraform infrastructure-as-code
- Operational runbook covering deployment, monitoring, and common issues
- Environment access handover (AWS, Docker Hub, DNS, secrets management)
- A knowledge-transfer walkthrough session with the sponsor